CLASS 9 · CBSE · GANITA MANJARI PART 1 · SHORT NOTES

Chapter 6: Measuring Space: Perimeter and Area

Topic-wise short notes with solved examples

This chapter measures the boundary and the region of plane shapes: the circumference and arcs of a circle, the areas of triangles and quadrilaterals (including Heron's and Brahmagupta's formulas), and the areas of circles, sectors and segments.

1 Perimeter and the number π

- The **perimeter** of a shape is the total length of its boundary. Square of side a: $4a$. Equilateral triangle of side a: $3a$. Rectangle with sides a and b: $2(a+b)$.
- For every square the ratio perimeter : side is $4:1$, whatever its size. In the same way, for every circle the ratio **circumference : diameter** is the same number. This number is called π.

$$\pi = \frac{C}{D} \qquad C = \pi d = 2\pi r$$

- π is **irrational**: its decimal form $3.14159265\ldots$ never ends and never repeats (proved by Lambert in 1761). So $\pi \approx \frac{22}{7}$ and $\pi \approx 3.14$, but $\pi \neq \frac{22}{7}$.

Who	Value or result for π
Mesopotamia (c. 1900 BCE)	$3\frac{1}{8} = 3.125$
Archimedes (c. 250 BCE)	$3\frac{10}{71} < \pi < 3\frac{1}{7}$
Zu Chongzhi (480 CE)	$\frac{22}{7}$ and $\frac{355}{113} \approx 3.1415929$
Āryabhaṭa (499 CE)	$\frac{62832}{20000} = 3.1416$, called "approximate"
Brahmagupta (628 CE)	$\sqrt{10} \approx 3.162$
Mādhava (14th century)	$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots$, an exact formula

REMEMBER

A regular hexagon drawn inside a circle of radius 1 has perimeter 6, which is less than the circumference 2π. This shows $\pi > 3$.

Example 1.1 A wheel has a diameter of 70 cm. How far does it move in one turn, and how many turns does it make in 1.1 km? Take $\pi = \frac{22}{7}$.

SOLUTION

One turn $=$ circumference $= \frac{22}{7} \times 70 = 220$ cm

1.1 km $= 110000$ cm, so the number of turns is $\frac{110000}{220}$.

220 cm; 500 turns

2 Length of an arc

- A semicircle is half the circle, so its length is πr. A quarter circle has length $\frac{\pi r}{2}$.
- An arc that subtends an angle θ° at the centre is the fraction $\frac{\theta}{360}$ of the circle:

$$\text{arc length} = \frac{\theta}{360} \times 2\pi r$$

- The **perimeter of a sector** is the arc length plus two radii.
- On a running track, a runner in an outer lane runs semicircles of larger radius. If the lanes are w metres apart, one full lap in the next lane is $2\pi w$ metres longer. The **stagger** at the start makes up for this.

Example 2.1 Find the length of an arc of a circle of radius 21 cm that subtends 120° at the centre. Take $\pi = \frac{22}{7}$.

SOLUTION

Arc $= \frac{120}{360} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{3} \times 132$

44 cm

Example 2.2 Find the perimeter of a sector of radius 14 cm and angle 90°. Take $\pi = \frac{22}{7}$.

SOLUTION

Arc $= \frac{90}{360} \times 2 \times \frac{22}{7} \times 14 = 22$ cm

Perimeter $= 22 + 14 + 14$

50 cm

Example 2.3 A semicircle is drawn on a segment PQ. Three smaller semicircles are drawn on three parts of PQ that together make up PQ. Which path from P to Q is longer?

SOLUTION

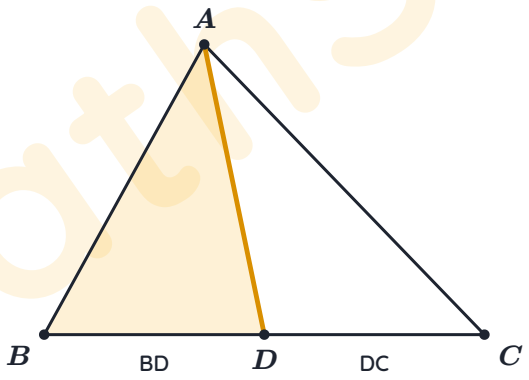
If the large radius is a and the small radii are b, c, d, then $2a = 2b + 2c + 2d$, so $a = b + c + d$.

Large path $= \pi a$. Small paths together $= \pi b + \pi c + \pi d = \pi(b + c + d)$.

The two paths are equal in length.

3 Area of rectangles, parallelograms and triangles

- The unit of area is a square of side 1 unit. Rectangle: ab. Square: a^2.
- **Parallelogram**: cutting a right-angled triangle from one end and moving it to the other turns it into a rectangle with the same base and height. So area = base $\times$ height $= bh$.
- The sides alone do not fix the area of a parallelogram. The height matters.
- **Triangle**: two congruent copies of a triangle fit together to make a parallelogram with the same base and height. So area $= \frac{1}{2} \times$ base $\times$ height.
- **Theorem**: a **median** of a triangle divides it into two triangles of equal area, because they have equal bases and the same height.



AD is a median. BD = DC, so the two triangles have equal area.

- **Trapezium** with parallel sides a, b and height h: area $= \frac{1}{2}(a + b)h$.
- **Rhombus or kite** with diagonals d_1 and d_2: area $= \frac{1}{2}d_1 d_2$.

Example 3.1 The parallel sides of an isosceles trapezium are 30 cm and 14 cm, and each of the other sides is 17 cm. Find its area.

SOLUTION

Dropping perpendiculars from the ends of the shorter side leaves $\frac{30 - 14}{2} = 8$ cm at each end of the longer side.

Height $= \sqrt{17^2 - 8^2} = 15$ cm. Area $= \frac{1}{2}(30 + 14) \times 15$.

330 cm²

4 Heron's formula

- For a triangle with sides a, b, c, first find the **semi-perimeter** $s = \dfrac{a+b+c}{2}$. Then

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

- For an **equilateral triangle** of side a this gives $\dfrac{\sqrt{3}}{4}a^2$, the same as half base times height with height $\dfrac{\sqrt{3}}{2}a$.
- Two more formulas: if R is the radius of the circumcircle and r the radius of the incircle, then Area $= \dfrac{abc}{4R}$ and Area $= r \times s$.

Example 4.1 Find the area of a triangle with sides 13 cm, 14 cm and 15 cm. Then find the radius of its incircle.

SOLUTION

$s = \dfrac{13+14+15}{2} = 21$

Area $= \sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} = 84$ cm²

Area $= r \times s$, so $r = \dfrac{84}{21}$.

Area 84 cm²; inradius 4 cm

Example 4.2 The sides of a triangular plot are in the ratio $3 : 4 : 5$ and its perimeter is 96 m. Find its area.

SOLUTION

The sides are $3k$, $4k$, $5k$ with $12k = 96$, so $k = 8$: the sides are 24 m, 32 m and 40 m.

$s = 48$. Area $= \sqrt{48 \times 24 \times 16 \times 8} = \sqrt{147456} = 384$

384 m² (check: it is right-angled, and $\dfrac{1}{2} \times 24 \times 32 = 384$)

5 Brahmagupta's formula for a cyclic quadrilateral

- The four sides alone do not fix the area of a quadrilateral: a rhombus of side 3 can be pushed flatter and flatter. One more fact is needed.
- If the quadrilateral is **cyclic**, with sides a, b, c, d and semi-perimeter $s = \dfrac{a+b+c+d}{2}$, then (Brahmagupta, 628 CE)

$$\text{Area} = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

- A triangle is the special case $d = 0$: the formula then becomes Heron's formula. So Brahmagupta's formula **generalises** Heron's formula.
- Check with a rectangle of sides a, b: $s = a + b$, and the formula gives $\sqrt{b \cdot a \cdot b \cdot a} = ab$.

Example 5.1 A cyclic quadrilateral has sides 7 cm, 24 cm, 20 cm and 15 cm. Find its area.

SOLUTION

$s = \dfrac{7+24+20+15}{2} = 33$

Area $= \sqrt{26 \times 9 \times 13 \times 18} = \sqrt{234 \times 234}$

234 cm²

6 Squaring a rectangle

- To **square** a shape means to construct a square with the same area.
- Baudhāyana's construction for a rectangle with sides a and b uses the identity

$$ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$$

- In a right-angled triangle with hypotenuse $\dfrac{a+b}{2}$ and one side $\dfrac{a-b}{2}$, the third side x has $x^2 = ab$. A square on that side has the same area as the rectangle.

Example 6.1 Find the side of the square equal in area to a rectangle 9 cm by 4 cm, using the identity.

SOLUTION

$\left(\dfrac{13}{2}\right)^2 - \left(\dfrac{5}{2}\right)^2 = \dfrac{169-25}{4} = 36$

The side is 6 cm.

7 Area of a circle

- For any shape, (perimeter)² : area stays the same when the shape is enlarged. For a circle the area is proportional to r^2, and Archimedes showed that the constant is exactly π.

$$A = \pi r^2 = \frac{1}{2} \times \text{circumference} \times \text{radius}$$

- Nīlakaṇṭha's picture: cut the circle into many thin slices and fit them together alternately. They form nearly a parallelogram with base πr (half the circumference) and height r, so the area is $\pi r \times r$.

Example 7.1 Find the area of a circle whose circumference is 44 cm. Take $\pi = \dfrac{22}{7}$.

SOLUTION

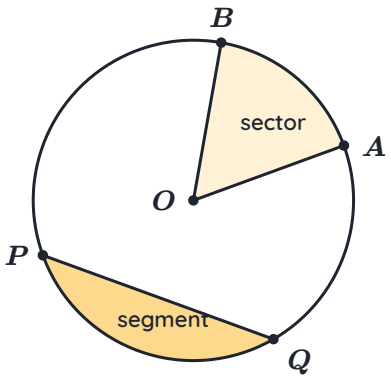
$2 \times \dfrac{22}{7} \times r = 44$, so $r = 7$ cm.

Area $= \dfrac{22}{7} \times 49$

154 cm²

8 Sector and segment of a circle

- A **sector** is the region between two radii and the arc joining their ends. A **segment** is the region between a chord and its arc.



$$\text{Area of a sector} = \frac{\theta}{360} \times \pi r^2$$

- Area of a **minor segment** = area of the sector − area of the triangle formed by the two radii and the chord.
- Area of the **major** sector or segment = area of the circle − area of the minor one.

Example 8.1 Find the area of a sector of radius 21 cm and angle $120°$. Take $\pi = \frac{22}{7}$.

SOLUTION

$$\text{Area} = \frac{120}{360} \times \frac{22}{7} \times 21 \times 21 = \frac{1}{3} \times 1386$$

462 cm²

Example 8.2 A chord of a circle of radius 14 cm subtends $90°$ at the centre. Find the area of the minor segment. Take $\pi = \frac{22}{7}$.

SOLUTION

$$\text{Sector} = \frac{90}{360} \times \frac{22}{7} \times 14 \times 14 = 154 \text{ cm}^2$$

$$\text{Triangle} = \frac{1}{2} \times 14 \times 14 = 98 \text{ cm}^2 \text{ (the two radii are at right angles)}$$

Segment $= 154 - 98 = 56$ **cm²**

Example 8.3 The minute hand of a clock is 14 cm long. Find the area it sweeps in 15 minutes.

SOLUTION

In 15 minutes the hand turns through $\frac{15}{60} \times 360^\circ = 90^\circ$.

Area $= \frac{90}{360} \times \frac{22}{7} \times 14 \times 14$

154 cm²

9 Common mistakes

- Using the diameter in place of the radius in πr^2 or $2\pi r$.
- Giving the arc length as the perimeter of a sector. Add the two radii.
- Forgetting to halve the perimeter before using Heron's or Brahmagupta's formula.
- Using Brahmagupta's formula for a quadrilateral that is not cyclic.
- Writing $\pi = \frac{22}{7}$ as an exact value. It is an approximation.
- Mixing units: change km to cm (or m) before dividing a distance by a circumference.

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